

Data mining final project(individual)

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Abstract

This individual final project focuses on participating in an online advertising bidding competition, where participants are required to predict bid prices for each day's test data. The winner is determined through a second-price auction mechanism, aiming to maximize click-through rate (CTR) and exposure under a limited budget.

The dataset provided is large-scale, containing millions of entries, and is highly imbalanced. In my approach, I first used a Random Forest model to rank feature importance and select the top features. These selected features were used to train XGBoost model for prediction. Additionally, I would continuously incorporate newly labeled data into the training set to improve performance over time.

Experimental results show that compared to Logistic Regression (LR), XGBoost, and deep learning models, the combination of Random Forest and XGBoost achieved the best AUC scores on the validation set. And I use it to predicting whether a click would occur.

This study demonstrates that even in the era dominated by deep learning, traditional machine learning methods not only offer better interpretability but can also deliver outstanding predictive performance.

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1 Introduction

In recent years, click-through rate (CTR) prediction has become a mainstream task in both industry and academia. How to increase CTR under a limited budget remains a major challenge in digital advertising. This project was conducted

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as part of a competition focused on real-time bidding for online advertisements. The objective was to develop a bidding agent trained on a large-scale, imbalanced dataset.

The main challenges included data imbalance, budget constraints, and model adaptability. To address these issues, I experimented with various prediction methods and evaluated their performance. Based on the results, I proposed an optimized model tailored for this competition and validated its effectiveness through actual bidding scenarios.

2 Dataset and Analysis

First, we conducted a preliminary analysis of the dataset. Using the pandas library, we observed that the dataset contains 1,760,309 records, with each record consisting of 22 features covering various categories. Some features are user-related, such as domain, user agent, and region etc, while others are related to the advertisement itself, such as ad slot format and ad slot width etc.

Upon inspecting the dataset, we found that most features are neither standard numerical nor categorical variables. Due to hashing or the presence of textual information, these features require substantial manual preprocessing and feature extraction. Therefore, we began by applying appropriate preprocessing techniques to each feature based on its characteristics.

2.1 Feature Extraction

We first process some features that need to be manually extracted, first of all, IP, user agent, user tags, time stamp. These features are not numerical or categorical features that can't directly used. We divide the IP into three segments according to commas to generate three different features, analyze the user agent, and encode the browser, operating system, mobile device and version accordingly to generate five different features. User tags are also divided into seven different features by comma segments, and for time stamp, both date and time features are extracted.

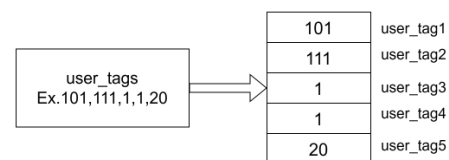


Figure 1. Example of extract new features in original features

2.2 One Hot Encoding

After analyzing the dataset, we found that most features are categorical, which aligns with the common characteristics of CTR prediction datasets—composed largely of high-dimensional and sparse categorical variables. Therefore, we applied one-hot encoding to these features. One-hot encoding is a classical technique for handling categorical variables, where a feature with N categories is transformed into an N -dimensional binary vector, with each category represented by a 0 or 1.

Although one-hot encoding can lead to a dimensionality explosion, especially when dealing with features that have a large number of categories, it more accurately captures the unordered nature of categorical data compared to other encoding methods such as label encoding, which may introduce misleading ordinal relationships. In this project, we carefully selected which features to one-hot encode in order to retain useful information while avoiding excessive dimensionality that could negatively impact model performance.

2.3 Remove part of features

After further analysis, we found that certain features, such as URL-related attributes, were extremely sparse and highly dispersed. Even the most frequently occurring values accounted for less than one percent of the entire dataset. Such characteristics may negatively impact the model's ability to generalize during the training process. As a result, we decided not to include these features in our model. Instead, we removed them to allow the model to focus on other, more informative features.

Finally, after statistics, we established a total of 528 dimensional features, as shown in the following figure:

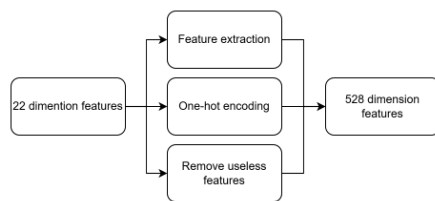


Figure 2. Data preprocessing work flow

3 Model and Methodology

For the established feature data set, we will use different model training methods to find the best fit. This part can be divided into two categories: Machine Learning and Deep Learning:

3.1 Machine Learning

3.1.1 Logistic Regression.

As a starting point, we used the simplest model—Logistic Regression (LR)—as our baseline. As one of the earliest algorithms developed for classification tasks, LR is known for its

simplicity, efficiency, and ease of parallelization. It is well-suited for quickly building an initial version of a model with acceptable performance.

However, Logistic Regression also has some obvious limitations. As a linear classification model, it assumes all features are independent, and it may fail to capture the interactions between features that could significantly affect prediction accuracy.

3.1.2 XGBoost. The second model we experimented with was XGBoost, an algorithm based on Gradient Boosted Decision Trees (GBDT). It is widely used in various data science competitions due to its high performance and flexibility. XGBoost builds decision trees iteratively, where each new tree focuses on correcting the errors made by the previous ones to minimize the overall loss function. It is highly efficient and effective, and it can also output feature importance scores, which greatly enhance model interpretability. In the Day 1 test data, XGBoost was used as the predictive model.

3.1.3 RandomForest+XGBoost.

The third method combines random forest for feature selection and XGBoost for classification. This attempt is to solve the problem that when we only use the XGBoost model, the model complexity will increase due to the high dimensionality, which affects the model's fitting performance and increases the training time. When performing CTR prediction tasks, a large number of high-dimensional features are its biggest feature, so I hope to reduce the complexity of the model itself and shorten the training time by combining the models, improve the model's score in the AUC indicator, and bring better bidding results.

3.2 Deep Learning

As the most popular prediction method nowadays, deep learning is also widely used in CTR prediction models. Deep learning's ability to fit nonlinear features and automatically learn high-order cross-information is considered to be an effective method for predicting click-through rate. Therefore, I also took the opportunity of the final report to implement a deep learning model with an Embedding layer + DNN architecture. Due to the introduction of the Embedding layer, I don't need to do too much processing on the input features. I only need to perform Label encoding on the category features, which greatly reduces the manpower part, which is also the attractive part of deep learning.

3.3 Online Updating Strategy

In order to increase the proportion of positive samples, I also added the data predicted as positive to the data for a new round of prediction after receiving the feedback file, hoping to get better results.

3.4 Second-price auction

The competition adopts a second-price auction (SPA) mechanism, in which the highest bidder wins, but pays the second highest price. This design encourages participants to bid their true valuations for each impression. On Day 2, I adopted an aggressive strategy by bidding a fixed amount of 350 units for all ads predicted to have a high probability of being clicked, reflecting strong confidence in the model's predictions. On day 3, I refined this approach by assigning bid prices in descending order according to the predicted click probability, aiming to optimize budget allocation and test the impact of a more granular probability-based bidding strategy.

4 Result

In the model and methodology selection, we mentioned the use of machine learning and deep learning methods to train the CTR prediction model. The following will show the training results and performance on the validation set and test set.

4.1 AUC score comparison

	Train AUC score	Vali AUC score
LR	0.8694	0.8664
XGBoost	0.9368	0.8238
RandomForest+XGBoost	0.9332	0.9031
Embedding+DNN	0.8254	0.6078

Table 1. AUC scores on the training set and validation set

In this chart,we can find that among the four different models, XGBoost and Embedding+DNN have serious overfitting problems. LR model can also achieve good results after appropriate artificial feature processing. And Randomforest+XGBoost has the best performance in validation set.

4.2 Recall value and FPR comparison

	Recall	FPR
LR	0.1726	0.0001
XGBoost	0.1547	0
RandomForest+XGBoost	0.4748	0.003
Embedding+DNN	1	1

Table 2. Recall and FPR rate of each model on validation set

In this chart,the DNN model achieved the highest Recall, successfully capturing most of the clicked samples. However, this came at the cost of an high FPR, which resulted

in bidding for many non-clicked impressions—an undesirable outcome given the constraint of limited budget.

On the other hand, both Logistic Regression and XGBoost maintained very low FPRs, minimizing unnecessary spending, but their low Recall scores indicated that many potential clicked samples were missed, reducing the opportunity for successful bidding.

RandomForest + XGBoost model provided a more balanced performance, with a significantly higher Recall compared to LR and XGBoost, while still keeping FPR at a relatively low level. This balance suggests that the model can identify more valuable impressions without excessively bidding on unclicked ones, making it the most effective model under budget constraints.

4.3 GridSearch for hyperparameter

We also optimized the hyperparameters of our main model and calculated three different combinations using the GridSearch package. In the end, we chose the combination with the highest AUC and a good recall value.

eta	0.1	0.1	0.1
max_dep	6	6	6
sub_sam	1	1	0.8
colsample	1	1	0.8
scale_pos	10	20	10
AUC	0.90079	0.89664	0.90312
Recall	0.4712	0.4856	0.4712

Table 3. Result of GridSearch for hyperparameter

4.4 Confusion matrix of main model on the validation set

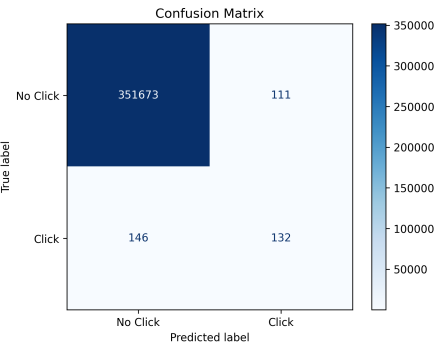


Figure 3. Confusion matrix of the RandomForest+XGBoost

The Confusion Matrix can better show that when we use the RandomForest+XGBoost model, we can distinguish the samples without clicks very well, and can also make a certain degree of prediction for the samples with clicks.

4.5 Day1 Predict Result

We used the XGBoost model in the Day 1 test set, but unfortunately we did not successfully bid for any ads.

4.6 Day2 Predict Result

	click	no click
AD	19	23

Table 4. Day2 bidding result

When we used RF+XGBoost model to predict the Day 2 test set, we successfully bid on 42 ads, 19 of which were clicked. According to the prompt, there are about 330 ads with clicks every day. On average, each contestant can bid on 8 ads with clicks, so successfully bidding on 19 ads is a very excellent result.

5 Conclusion

In summary, this competition fully demonstrates the specificity of CTR prediction in many data mining tasks. Large and extremely unbalanced data often causes general models to overfit or even make extreme predictions in pursuit of accuracy. This is true for both machine learning and deep learning. When we cannot properly process the data or modify the model, we often lose the part we really want to focus on.

In this era of deep learning, human-powered feature engineering seems to be a thing of the past. We rely heavily on deep learning to automatically select parameters when passing them. Although the power of computing is now developed, it can greatly reduce the loss of manpower, but when we focus on certain samples instead of overall accuracy, it may bring unsatisfactory results. The obvious shortcomings can be seen from the results of this experiment.

Machine learning provides us with a statistically based algorithm that obtains results in a rigorous way. The most important thing is that it can provide good interpretability for the model, allowing us to adjust our model through a clear context and get the results we want.

The development of deep learning has brought remarkable progress to human technology, but we should not ignore the power of artificial feature engineering and machine learning. Especially in situations where the number of samples is insufficient, model interpretability is high, or domain knowledge is required, machine learning combined with carefully designed feature engineering can often show stability and predictive capabilities that are not inferior to deep learning. This implementation also verified this point. Through appropriate data pre-processing and model combination, we successfully achieved satisfactory results on extremely unbalanced data sets, showing that data-oriented thinking and technology are still the key to model performance.